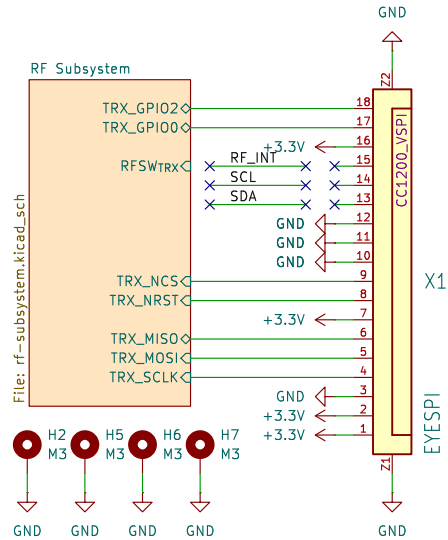
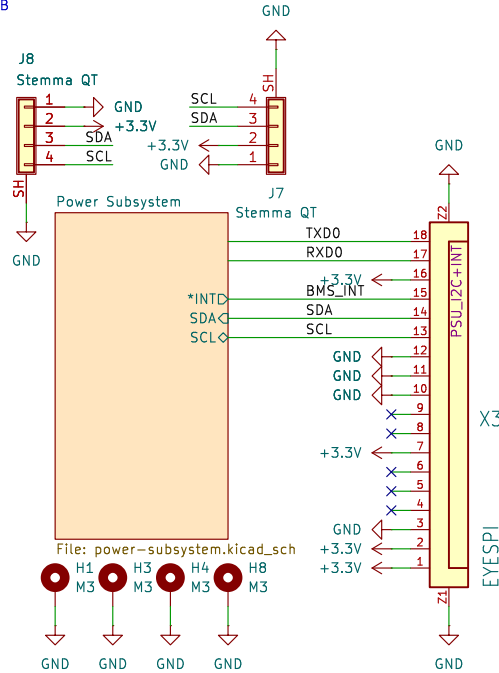


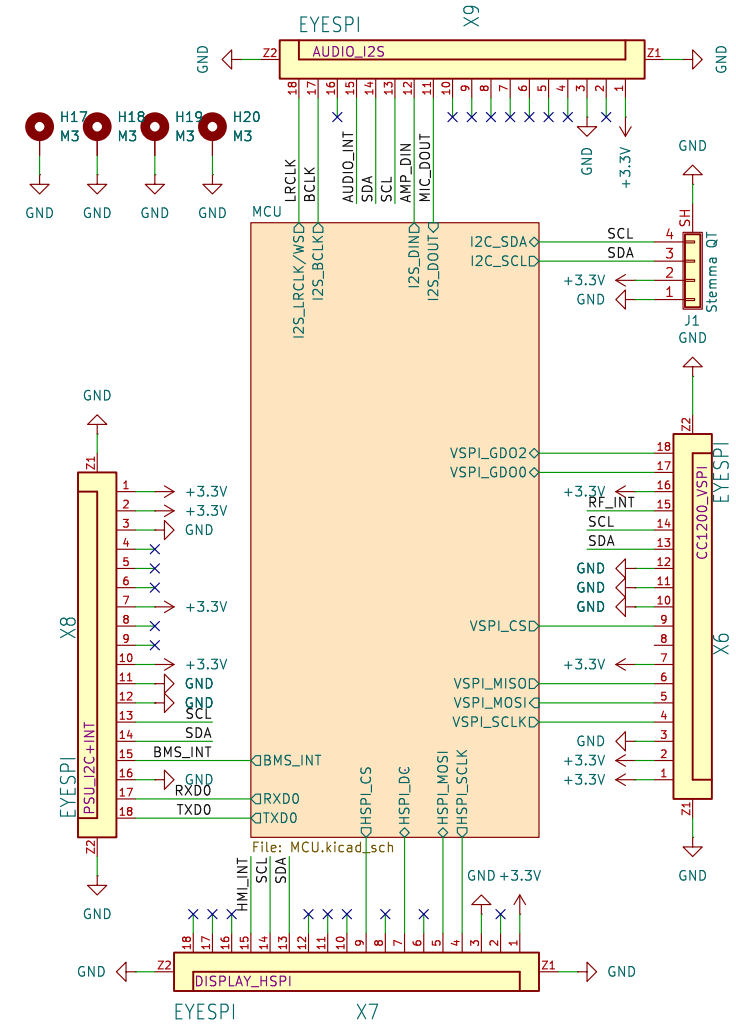
RF Modulino
 CC1200 on VSP1
 Unused I2C0+INT
 RX Switch on ???
 4x 3.3V return paths



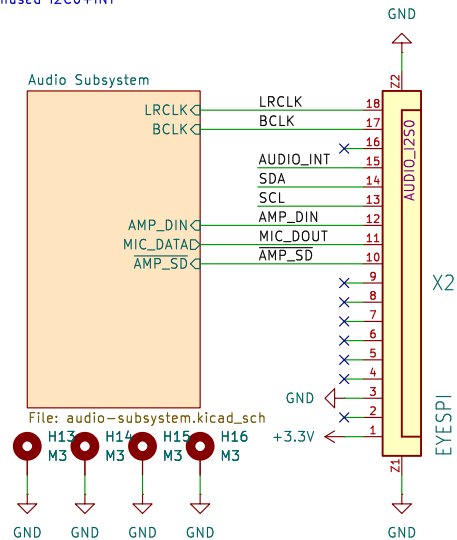
Battery Modulino
 BMS on I2C0 @ 0x6B



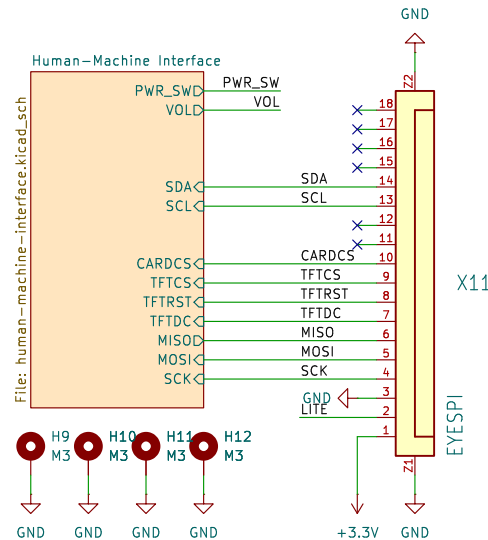
ESP32 Modulino
 X? EYESPI to Modulino Nodes



Audio Modulino
 Mic+AMP on I2S0:
 MAX98357A I036 INPUT
 ICS-43434 I033 OUTPUT
 Unused I2C0+INT



Human-Machine Interface Modulino
 Display on HSPI:
 SD Card Unused
 Port Expander on I2C0+INT @ ???



Sheet: /
 File: system-testbed.kicad_sch

Title:

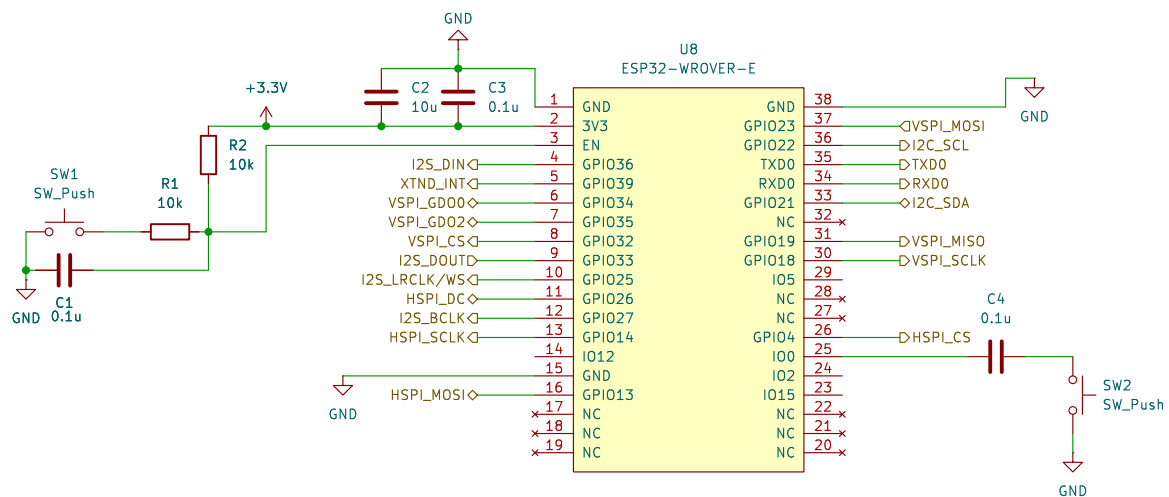
Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 1/10



Sheet: /MCU/
File: MCU.kicad_sch

Title:

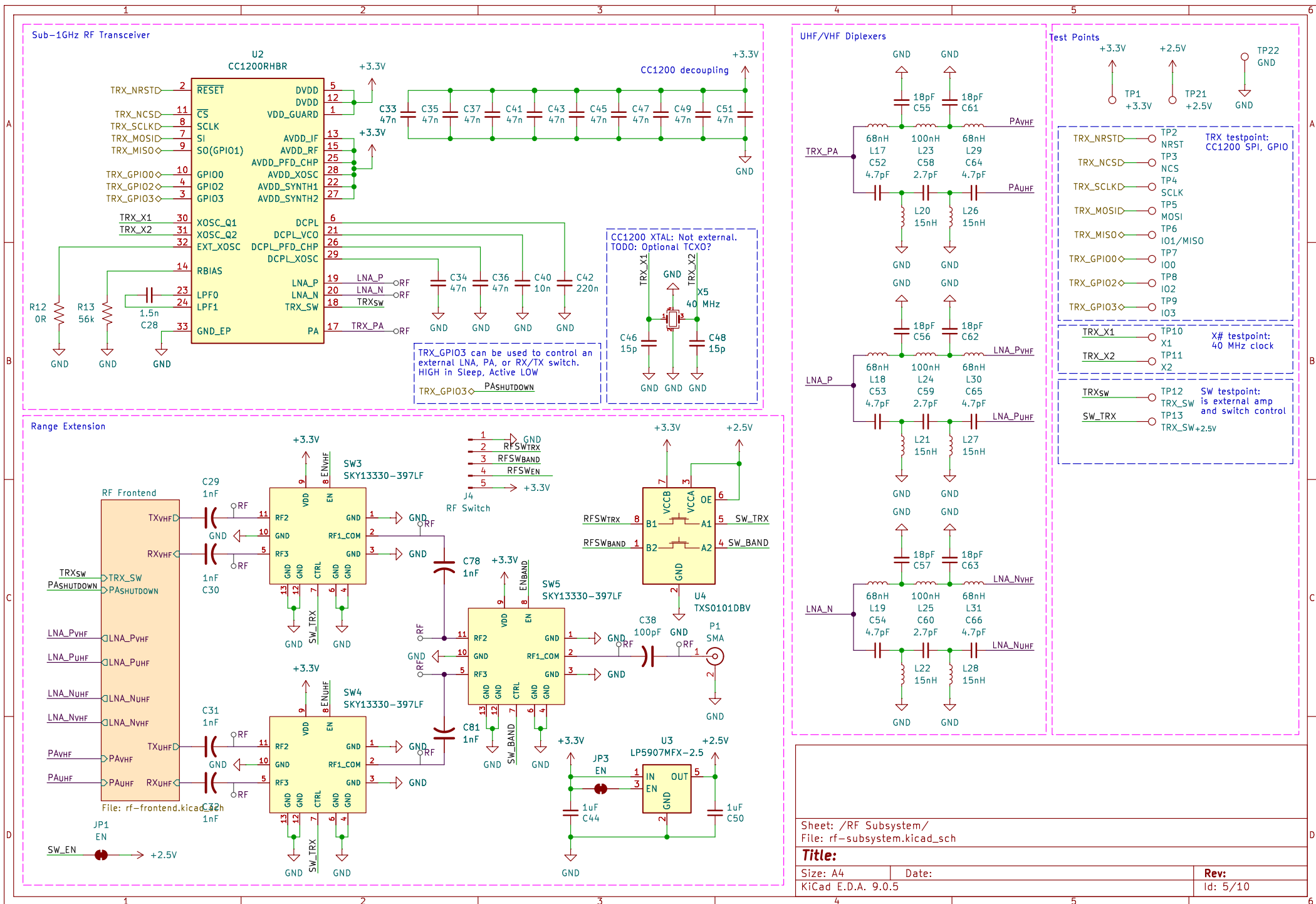
Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 2/10

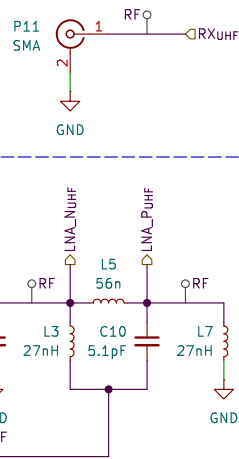


RF Frontend for UHF Reference 420-470 MHz Tune Separated RX and TX Signal Chains

RX Signal Chain

Single-ended External LNA into Balun for differential input of Internal LNA

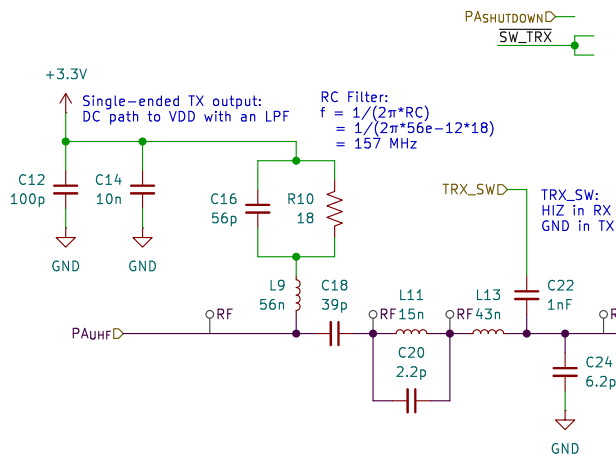
Balun as an LC Tank:
HIZ at $f = 429.9$ Mhz
(wL/wC)/($jwL-j/wC$) -> INF



TX Signal Chain

+3.3V

Single-ended TX output:
DC path to VDD with an LPF
RC Filter:
 $f = 1/(2\pi*RC)$
 $= 1/(2\pi*56e-12*18)$
 $= 157$ MHz



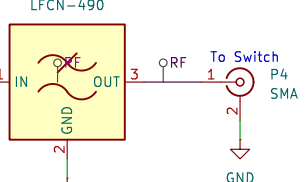
J2
Conn_01x06_Pin
2
3
4
5
6

From RF Modulino
P3 SMA
1
2

UHF 6W PA
Vcc2
Vcc1
VENABLE2
VENABLE1
VSHUTDOWN

File: rf-power-amplifier-uhf.kicad_sch

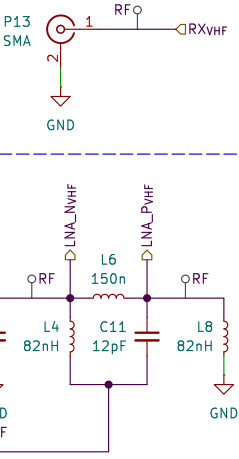
FL1
LFCN-490



RF Frontend for VHF Reference 169-170 MHz Tune Separated RX and TX Signal Chains

RX Signal Chain

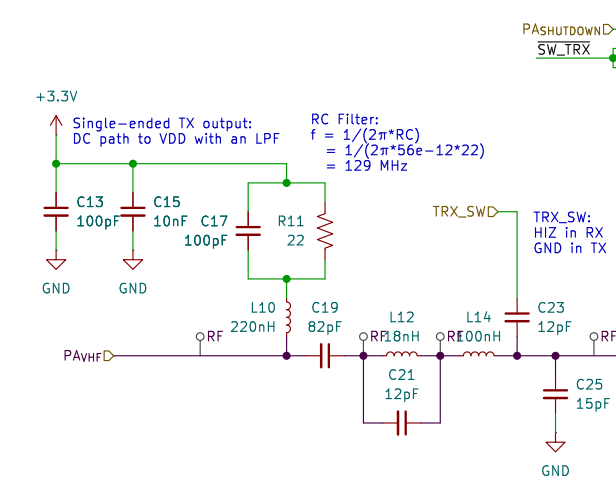
Single-ended External LNA into Balun for differential input of Internal LNA



TX Signal Chain

+3.3V

Single-ended TX output:
DC path to VDD with an LPF
RC Filter:
 $f = 1/(2\pi*RC)$
 $= 1/(2\pi*56e-12*22)$
 $= 129$ MHz



PASHUTDOWN
SW_TRX

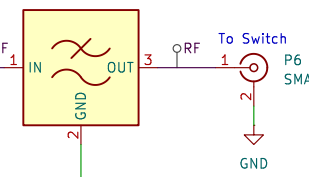
J3
Conn_01x06_Pin
2
3
4
5
6

P5 SMA
1
2

VHF 6W PA
Vcc2
Vcc1
VENABLE2
VENABLE1
VSHUTDOWN

File: rf-power-amplifier-vhf.kicad_sch

FL2
LFCN-160



Sheet: /RF Subsystem/RF Frontend/
File: rf-frontend.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.5

Date:

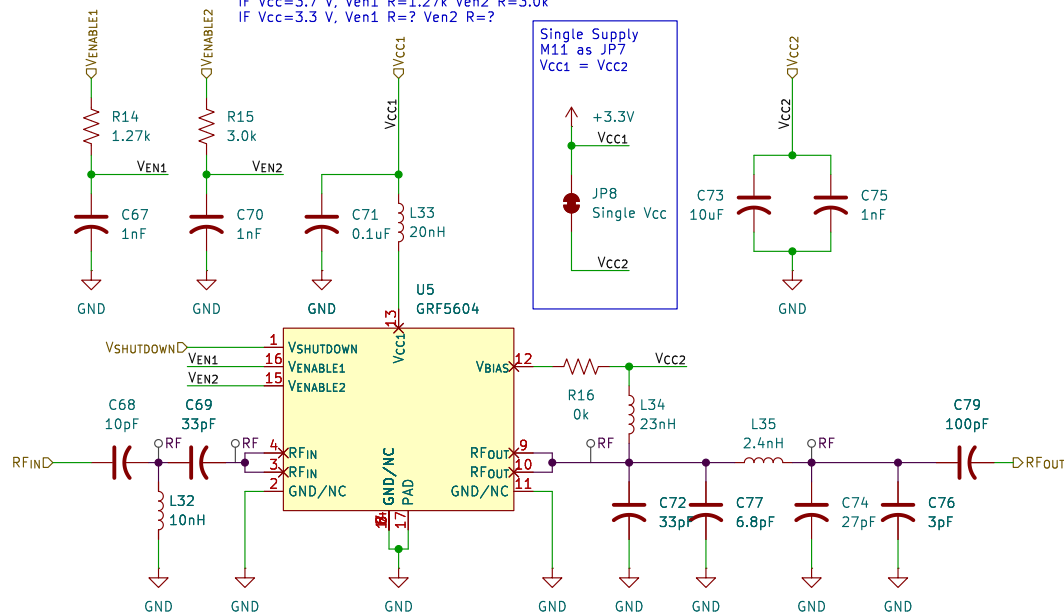
Rev:

Id: 4/10

RF Amplifier for UHF Reference 397-470 MHz Tune

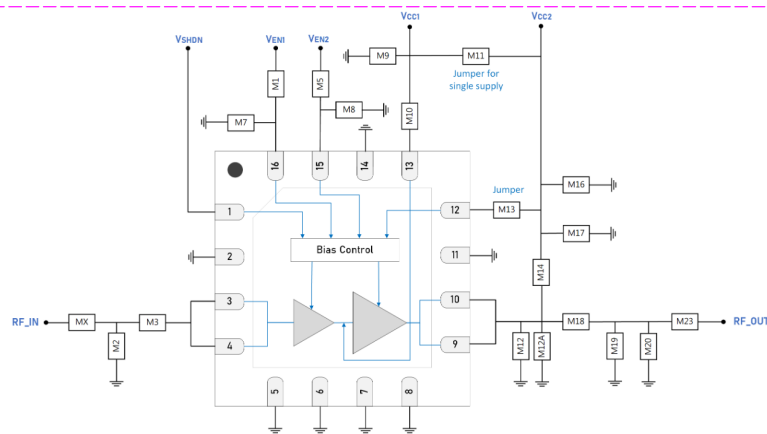
IF Vcc=5.0 V, Ven1 R=1.96k Ven2 R=6.04k
IF Vcc=3.7 V, Ven1 R=1.27k Ven2 R=3.0k
IF Vcc=3.3 V, Ven1 R=? Ven2 R=?

Single Supply
M11 as JP7
Vcc1 = Vcc2



Ref	Eval. Ref	450-470 MHz
R	M1	1.96k
L1	M2	10nH
C5	M3	33pF
C3	MX	10pF
R	M5	6.04k
C	M7	1000pF
C	M8	1000pF
C	M9	0.1uF
L	M10	20nH
R	M11	0k
C7	M12	33pF
C	M12A	2.0pF
R	M13	0k
L	M14	23nH
C	M16	**10uF
C	M17	1000pF
L	M18	2.2nH
C8	M19	27pF
C	M20	1.8pF
C	M23	100pF

**10 μ F must be rated for > 5 V at maximum ambient temperature. Manufacturer Part Number in this case = GRM155C80J106ME11D.



GRF5604 Standard Evaluation Board Schematic

Sheet: /RF Subsystem/RF Frontend/UHF 6W PA/
File: rf-power-amplifier-uhf.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.5

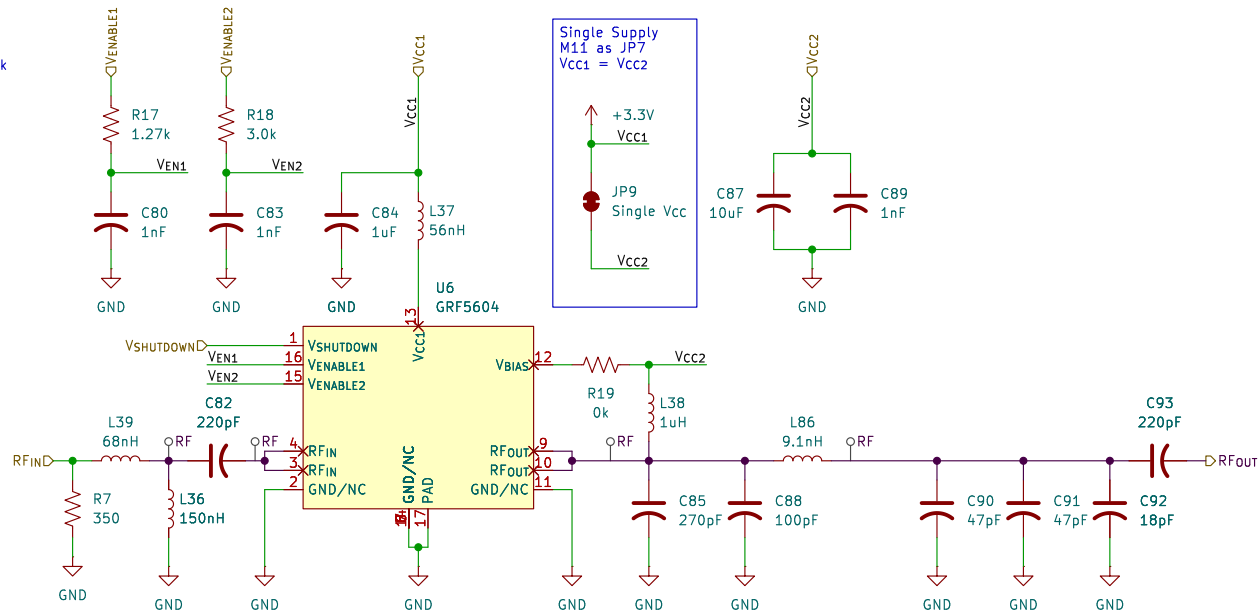
Date:

Rev:

Id: 6/10

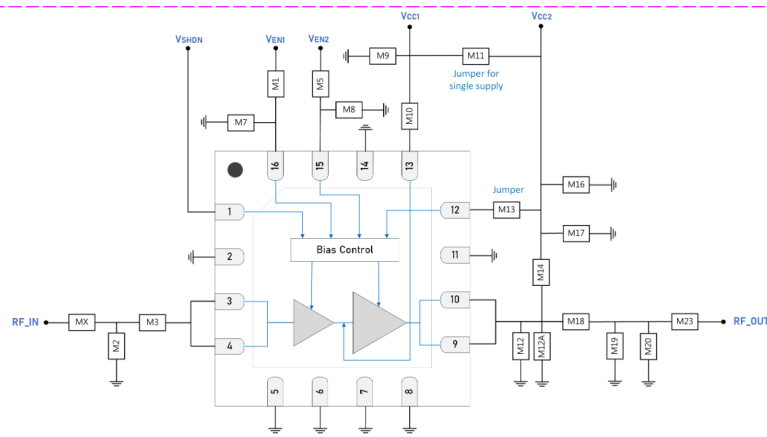
RF Amplifier for VHF Reference 135-175 MHz Tune

IF Vcc=5.0 V, Ven1 R=1.96k Ven2 R=6.04k
IF Vcc=3.7 V, Ven1 R=1.27k Ven2 R=3.0k
IF Vcc=3.3 V, Ven1 R=? Ven2 R=?



Ref	Eval. Ref	450-470 MHz
R	M1	1.96k
L1	M2	10nH
C5	M3	33pF
C3	MX	10pF
R	M5	6.04k
C	M7	1000pF
C	M8	1000pF
C	M9	0.1uF
L	M10	20nH
R	M11	0k
C7	M12	33pF
C	M12A	2.0pF
R	M13	0k
L	M14	23nH
C	M16	**10uF
C	M17	1000pF
L	M18	2.2nH
C8	M19	27pF
C	M20	1.8pF
C	M23	100pF

**10 uF must be rated for > 5 V at maximum ambient temperature. Manufacturer Part Number in this case = GRM155C80J106ME11D.



GRF5604 Standard Evaluation Board Schematic

Sheet: /RF Subsystem/RF Frontend/VHF 6W PA/
File: rf-power-amplifier-vhf.kicad_sch

Title:

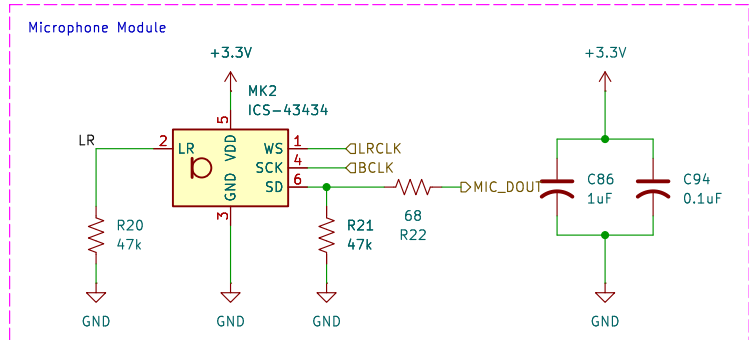
Size: A4

Date:

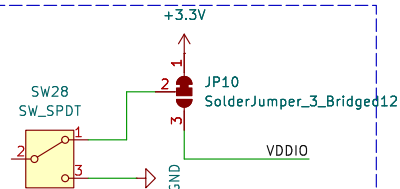
KiCad E.D.A. 9.0.5

Rev:

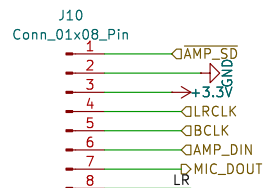
Id: 7/10



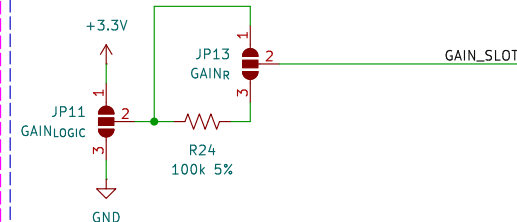
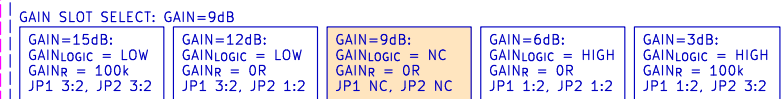
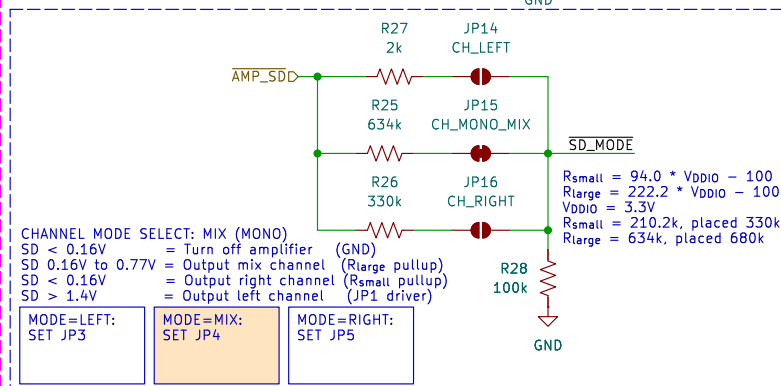
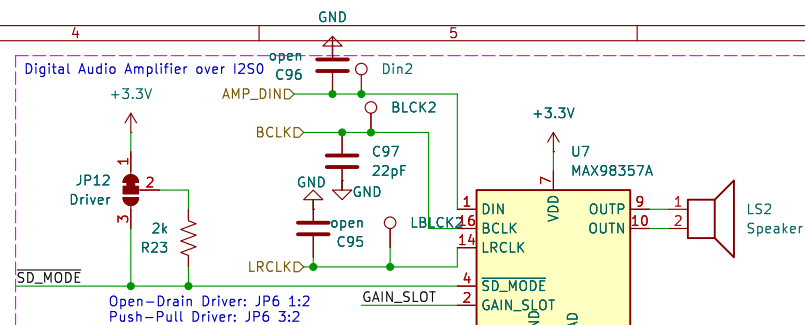
Shut Down Mode



BreadboardConnector



LR-In case of two micro modulino



Sheet: /Audio Subsystem/
File: audio-subsystem.kicad_sch

Title:

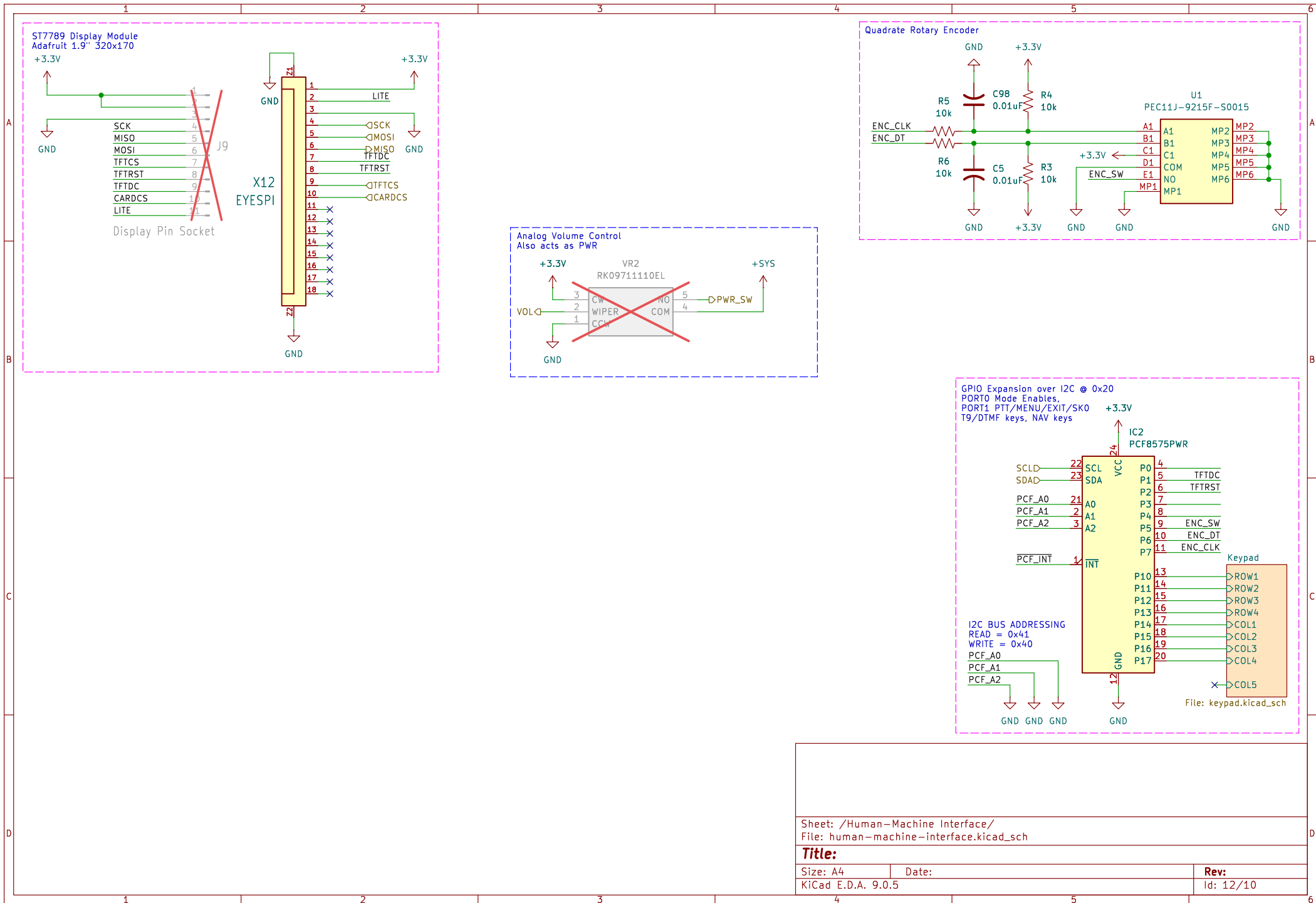
Size: A4

Date:

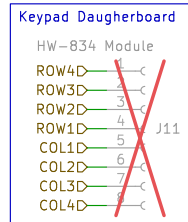
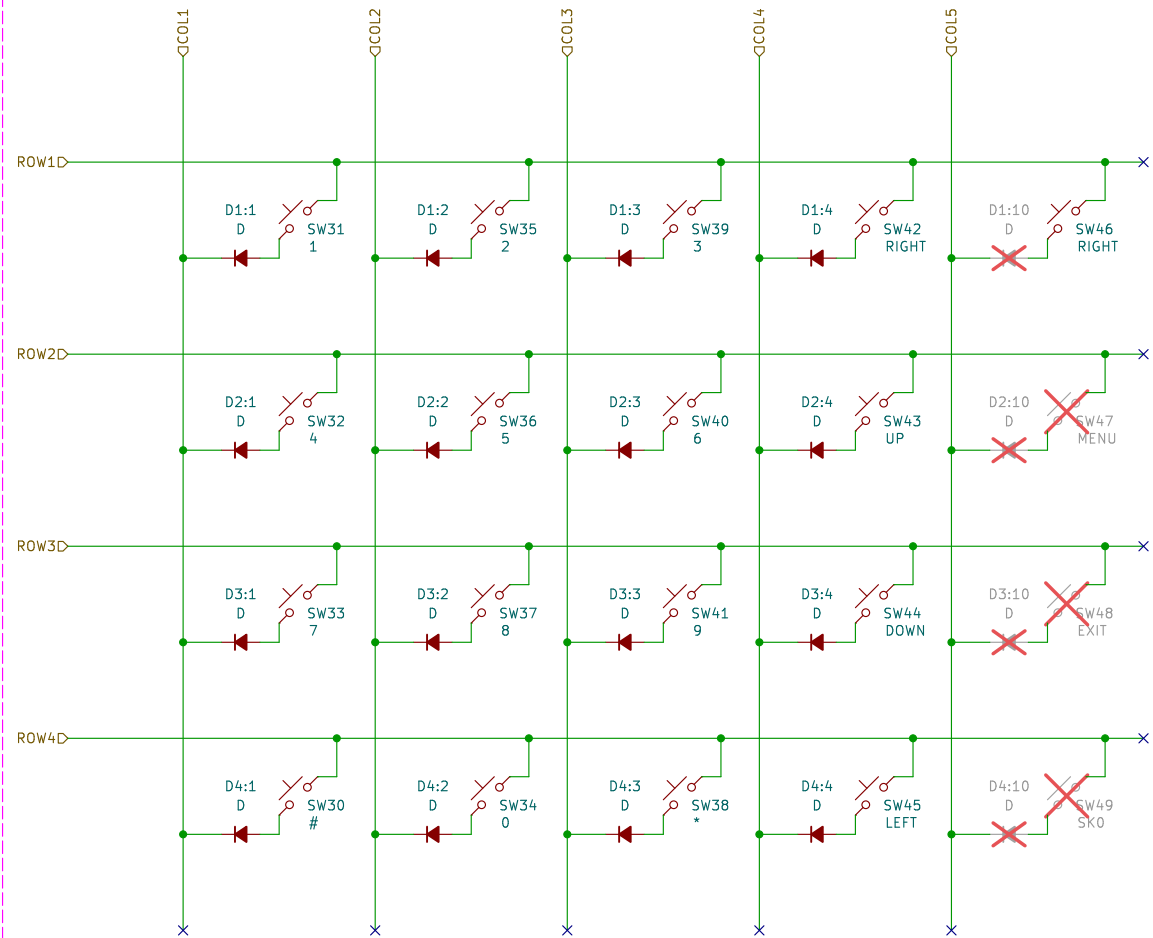
KiCad E.D.A. 9.0.5

Rev:

Id: 8/10



Keypad Matrix
 Application Note: The ROW pins get pulled up by the GPIO expander, with one ROW being pulled down.
 The scanner relies on the COL pins being read in as a logic swing to low.
 Remarked as Domino Logic, a rail-to-rail logic swing occurs.



Sheet: /Human-Machine Interface/Keypad/
 File: keypad.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 13/10